

```
In [1]: import numpy as np
a=np.array([[1,2,4],[5,8,7]])
print(a)
```

```
[[1 2 4]
 [5 8 7]]
```

```
In [2]: b=np.zeros((3,4))
print(b)
```

```
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
```

```
In [3]: b=np.zeros((4,4))
print(b)
```

```
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
```

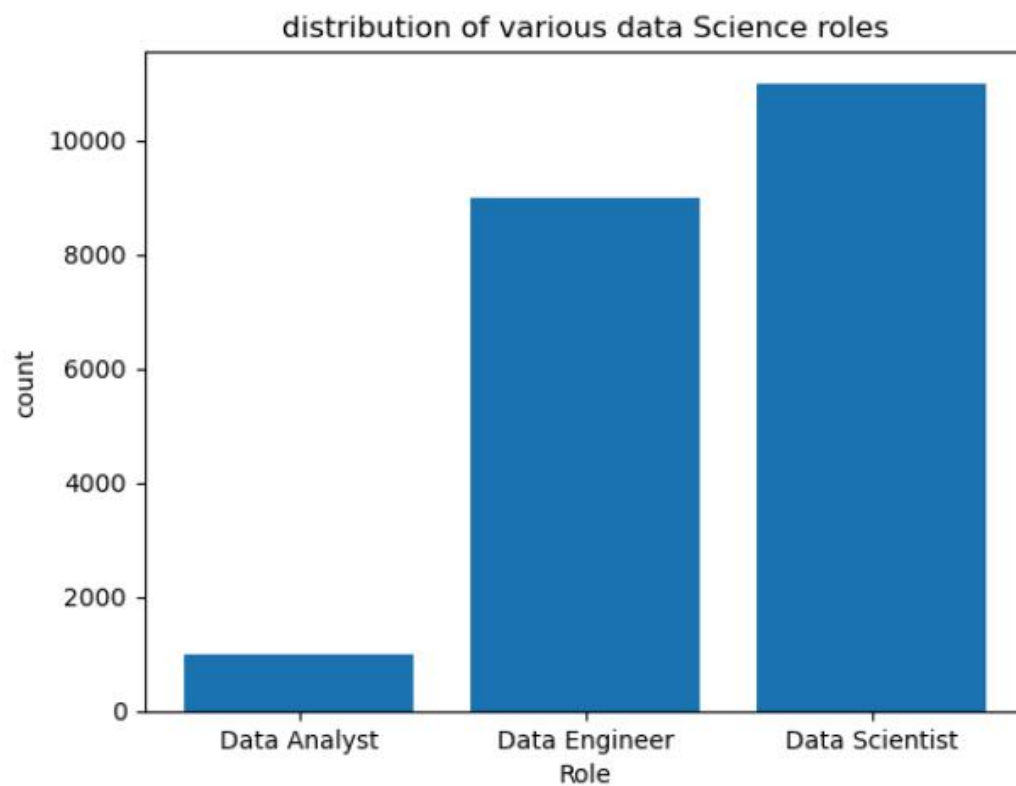
```
In [9]: c=np.full((3,3),6)
print(c)
```

```
[[6 6 6]
 [6 6 6]
 [6 6 6]]
```

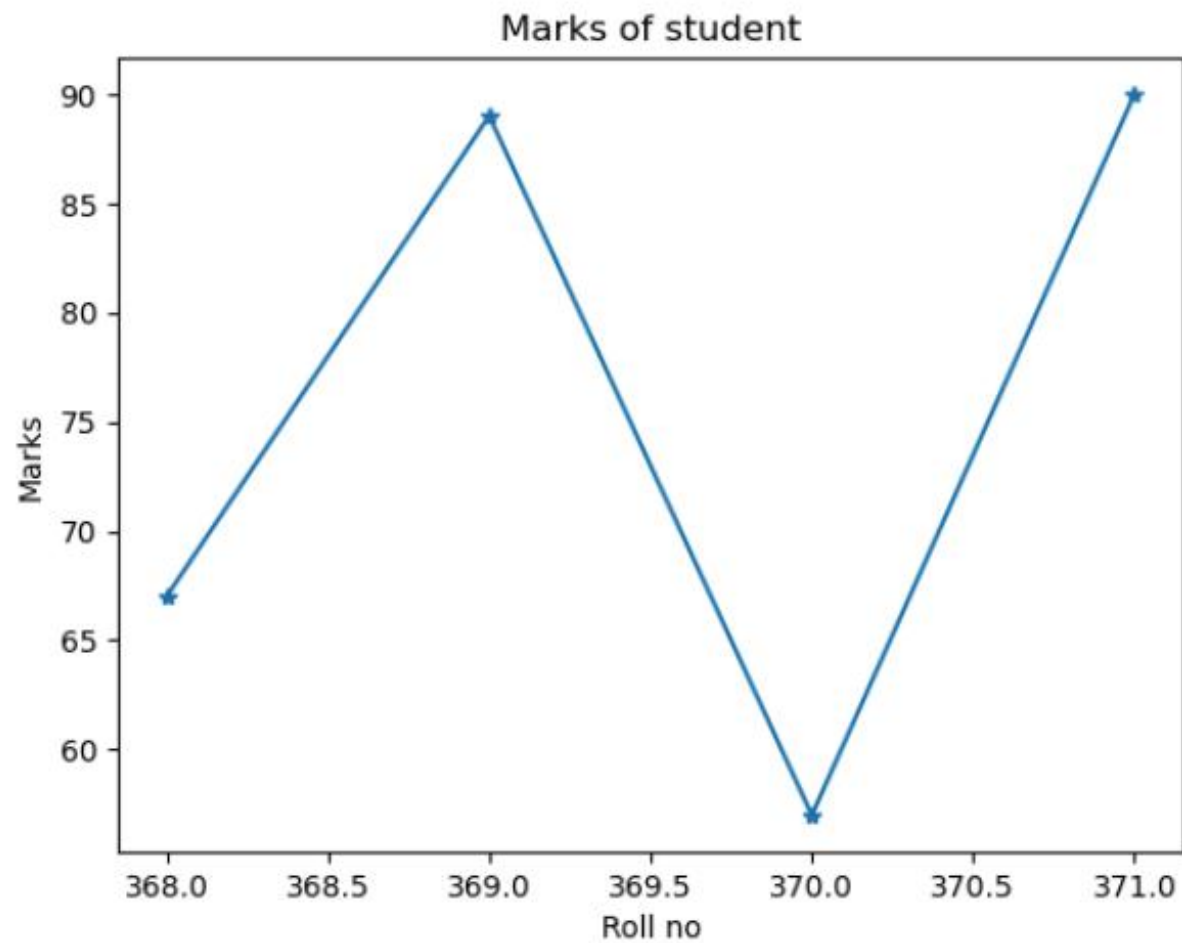
```
In [5]: d=np.random.random((2,4))
print(d)
```

```
[[0.13590369 0.20560214 0.5020129 0.05712532]
 [0.35391643 0.43280245 0.75022576 0.8190716 ]]
```

```
In [13]: import pandas as pd
import matplotlib.pyplot as plt
data = {'Role':['Data Analyst','Data Engineer','Data Scientist'], 'Job Posting':[1000,9000,11000]}
df=pd.DataFrame(data)
plt.bar(df['Role'],df['Job Posting'])
plt.title('distribution of various data Science roles')
plt.xlabel('Role')
plt.ylabel('count')
plt.show()
```

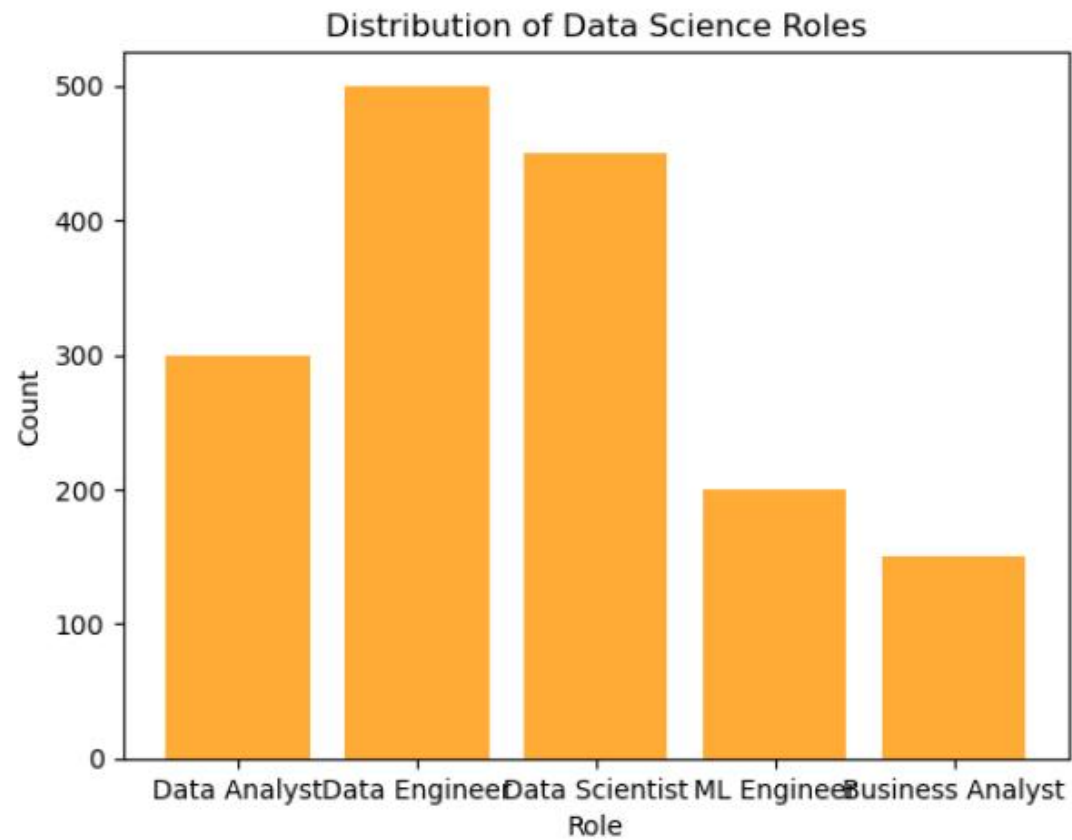


```
In [43]: import pandas as pd
import matplotlib.pyplot as plt
data={'Roll no':list(range(368,372)),
'Marks':[67,89,57,90]}
df=pd.DataFrame(data)
plt.plot(df['Roll no'],df['Marks'],marker='*')
plt.title('Marks of student')
plt.xlabel('Roll no')
plt.ylabel('Marks')
plt.show()
```

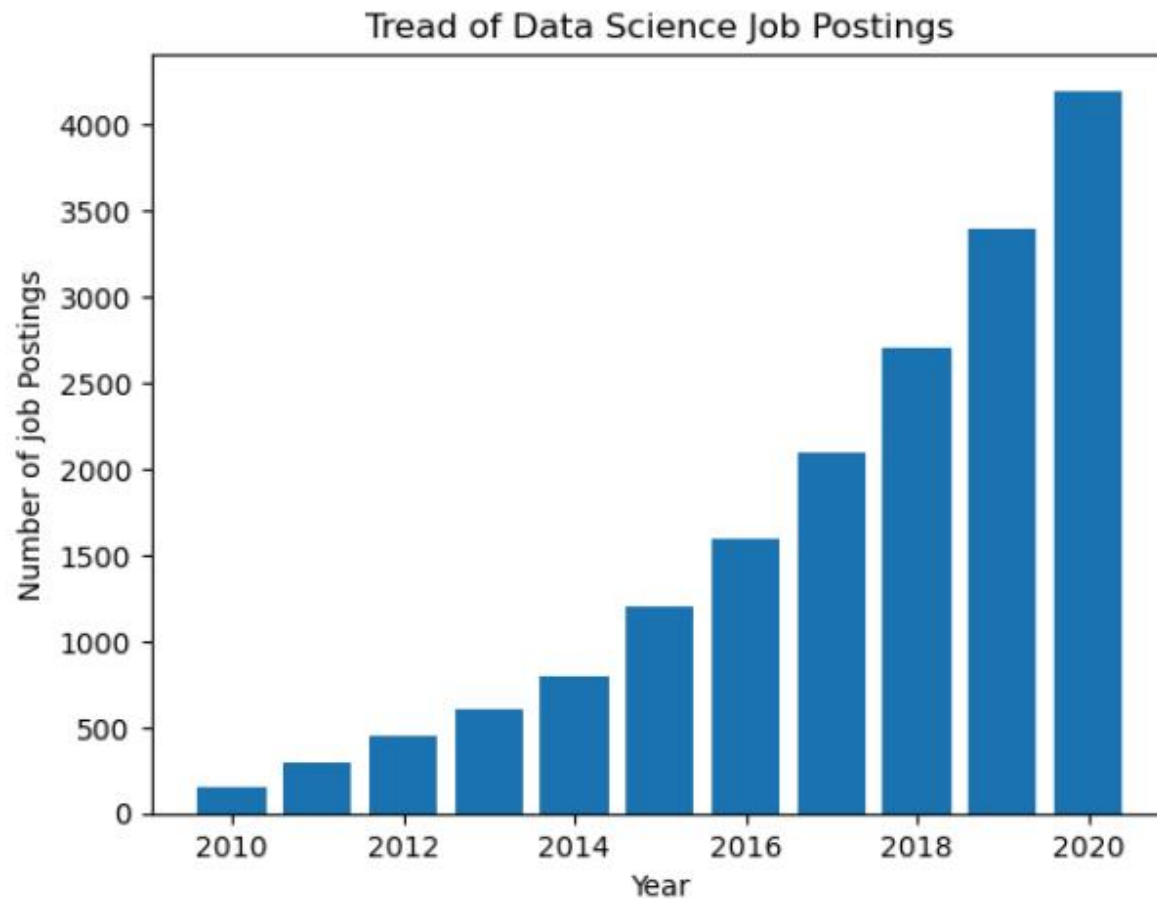


Year

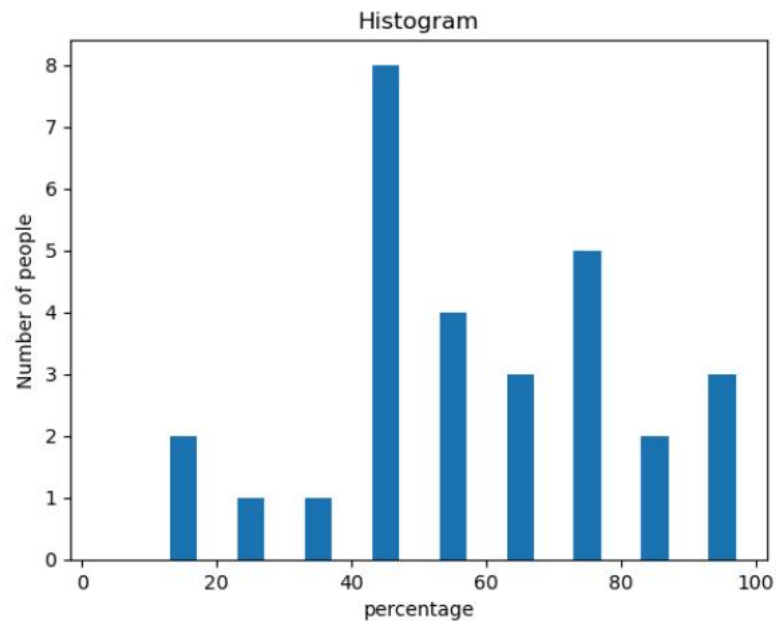
```
In [41]: import pandas as pd
import matplotlib.pyplot as plt
data={'Roles':['Data Analyst','Data Engineer','Data Scientist','ML Engineer','Business Analyst'],
      'Counts':[300,500,450,200,150]}
plt.bar(df['Roles'],df['Counts'],color='orange')
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```



```
In [26]: import pandas as pd
import matplotlib.pyplot as plt
data={'Year':list(range(2010,2021)),
'Job Postings':[150,300,450,600,800,1200,1600,2100,2700,3400,4200]}
df=pd.DataFrame(data)
plt.bar(df['Year'],df['Job Postings'])
plt.title('Tread of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of job Postings')
plt.show()
```



```
In [2]: from matplotlib import pyplot as plt
from matplotlib import pyplot as plt
percentage = [97,54,45,10, 20, 10, 30,97,50,71,40,49,40,74,95,80,65,82,70,65,55,70,75,60,52,44,43,42,45]
number_of_student = [0,10,20,30,40,50,60,70,80,90,100]
plt.hist(percentge, number_of_student, histtype='bar', rwidth=0.4
)
plt.xlabel('percentage')
plt.ylabel('Number of people')
plt.title('Histogram')
plt.show()
```



```
In [ ]: import matplotlib.pyplot as plt
plt([1,2,3][2,5,4])
plt.show
```



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```
In [20]: import pandas as pd
import matplotlib.pyplot as plt
x=[1,7,2]
y=pd.Series(x,index=["x","y","z"])
print(y)
```

```
x    1
y    7
z    2
dtype: int64
```

```
In [21]: x={"cars":["bmw","Audi"],"passings":[3,7]}
print(pd.DataFrame(x))
```

```
   cars  passings
0  bmw         3
1  Audi         7
```

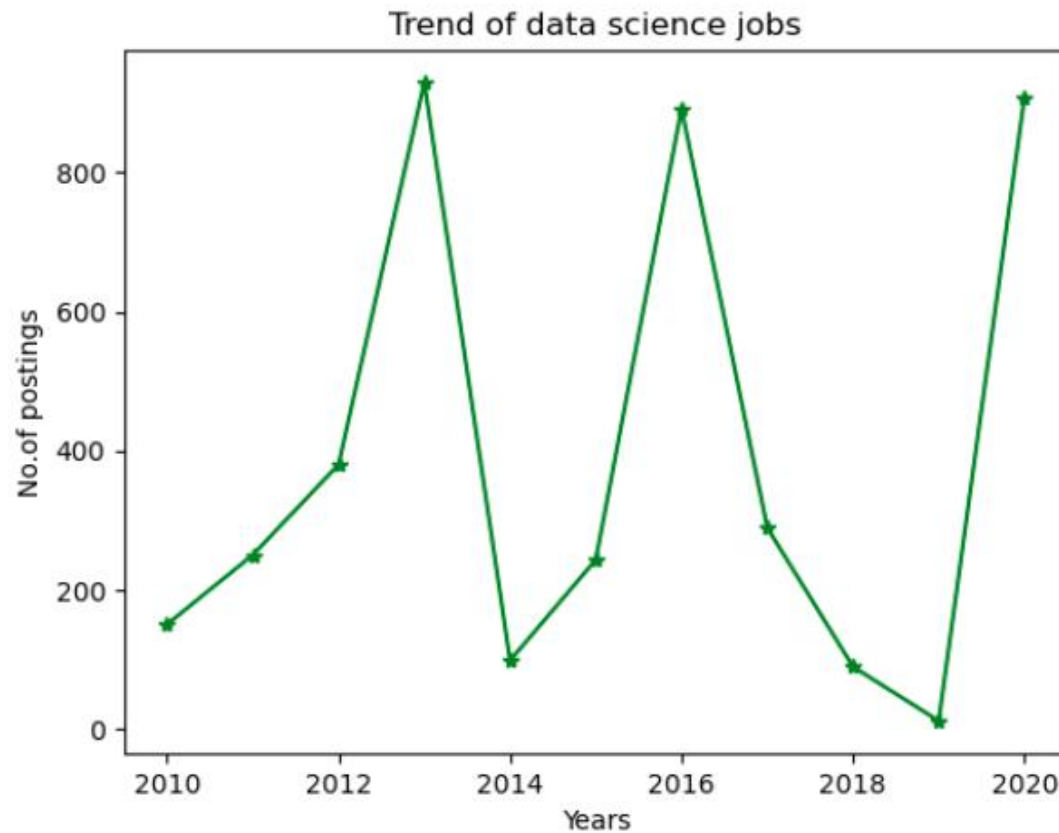
```
In [22]: x={"Subject":["Maths","CA","physics","Java","FDS"],"CAT1":[92,93,90,100,95],"CAT2":[96,87,98,95,87],"SEM":["0","A+","A","0","0"]}
y=pd.DataFrame(x,index=[1,2,3,4,5])
print(y)
```

	Subject	CAT1	CAT2	SEM
1	Maths	92	96	0
2	CA	93	87	A+
3	physics	90	98	A
4	Java	100	95	0
5	FDS	95	87	0

```
In [23]: print(pd.read_csv("studentdetails.csv"))
```

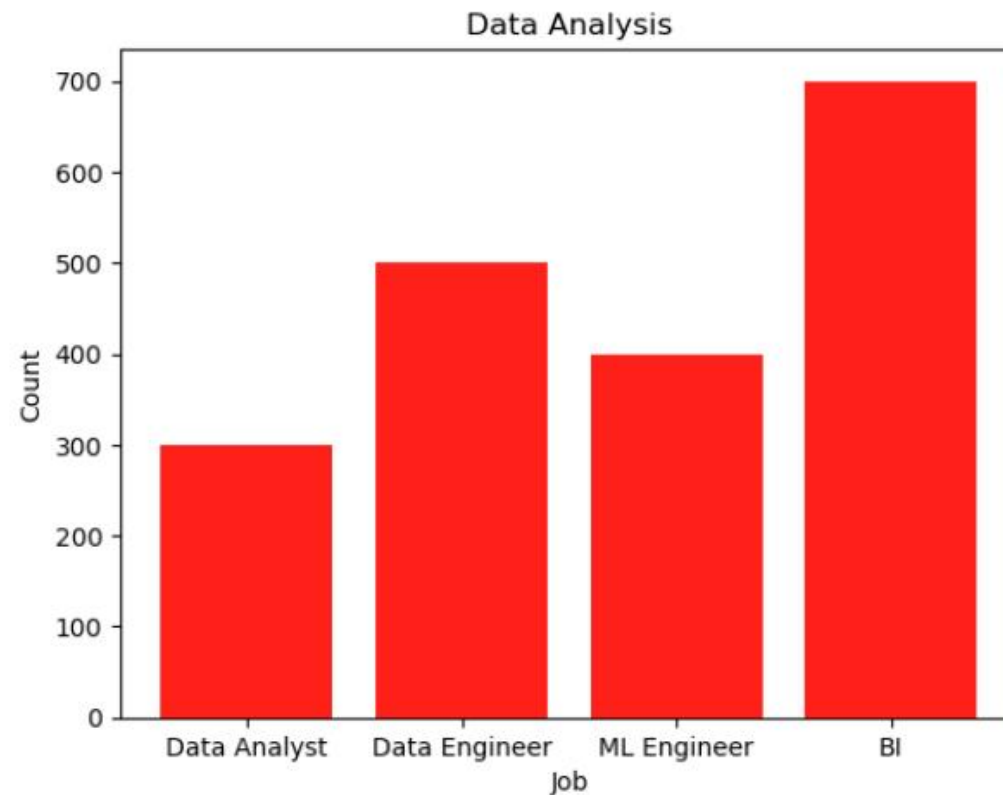
	St_id	St_Name	St_mark
0	101	hema	89
1	102	neeta	86
2	103	tina	92
3	104	hina	89
4	105	ram	78
5	106	arun	78
6	107	avana	78
7	108	shwetha	95
8	109	priya	78
9	110	latha	63

```
In [40]: import pandas as pd
import matplotlib.pyplot as plt
data={"years":list(range(2010,2021)),"jobpost":[150,250,380,930,98,243,892,289,89,11,909]}
a=pd.DataFrame(data)
plt.plot(a["years"],a["jobpost"],marker="*",color="Green")
plt.title("Trend of data science jobs")
plt.xlabel("Years")
plt.ylabel("No.of postings")
plt.show()
```



```
In [38]: role=["Data Analyst","Data Engineer","ML Engineer","BI"]
count=[300,500,400,700]
plt.bar(role,count,color="Red")
plt.title("Data Analysis")
plt.xlabel("Job")
plt.ylabel("Count")
plt.show();
```

```
In [38]: role=["Data Analyst","Data Engineer","ML Engineer","BI"]
count=[300,500,400,700]
plt.bar(role,count,color="Red")
plt.title("Data Analysis")
plt.xlabel("Job")
plt.ylabel("Count")
plt.show();
```



```
In [41]: print(pd.read_csv("dataroles.csv"))
```

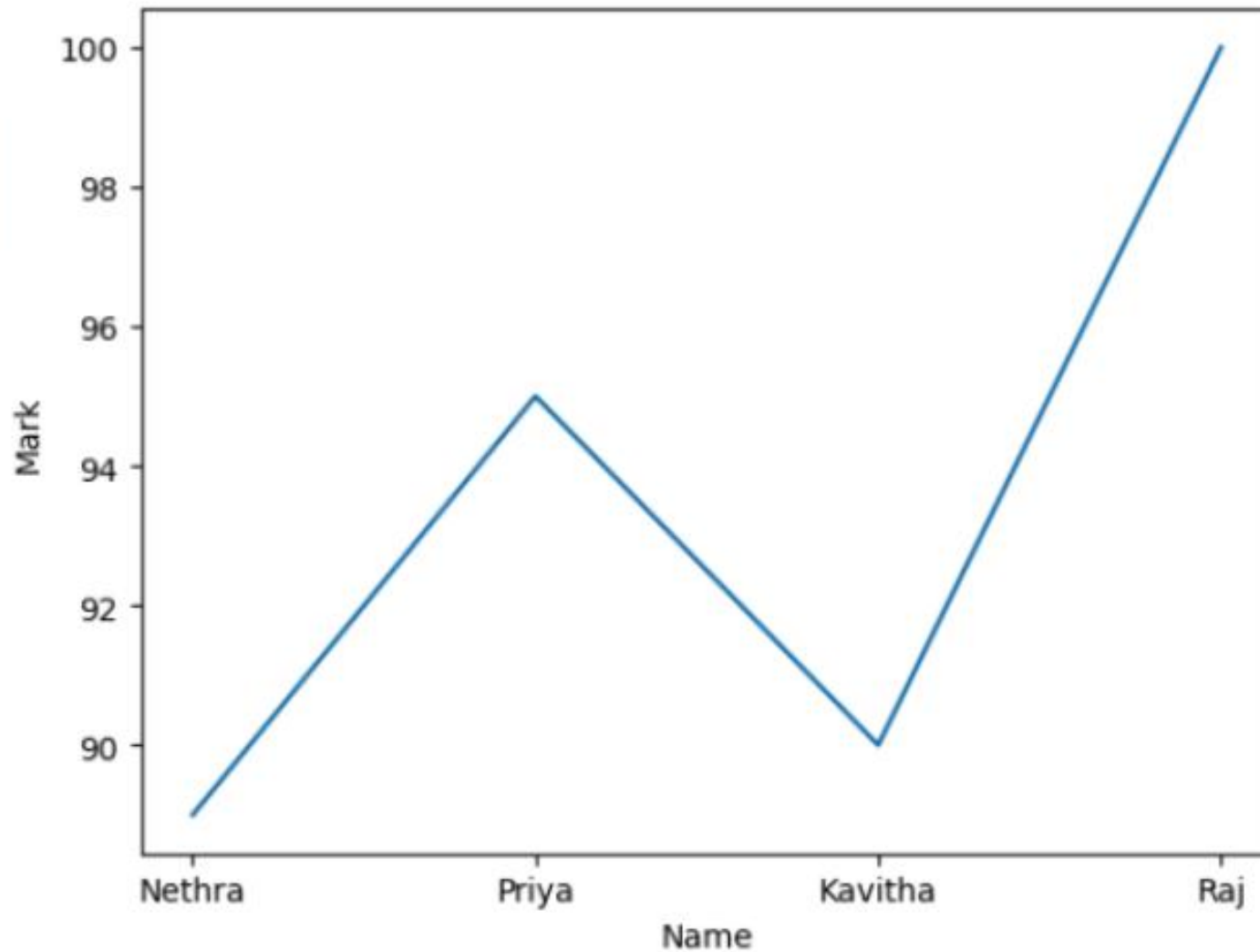
	ID	Name	Role	count
0	101	Janai	Data Analyst	89
1	102	Latha	Data Scientist	633
2	103	Shwetha	Data Engineer	800
3	104	Anu	Business Analyst	200
4	105	Gayathri	ML Engineer	898



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```
import pandas as pd
import matplotlib.pyplot as plt
data={"Name":["Nethra","Priya","Kavitha","Raj"],
      "Mark":[89,95,90,100]}
df=pd.DataFrame(data)
plt.plot(df["Name"],df["Mark"])
plt.xlabel("Name")
plt.ylabel("Mark")
plt.show()
```



```
In [5]: import pandas as pd
structured_data=pd.DataFrame({
    'Id':[12,13,14],
    'Name':['raj','jhon','kumar']
})
print(structured_data)
```

	Id	Name
0	12	raj
1	13	jhon
2	14	kumar

```
In [9]: import pandas as pd
structured_data=pd.DataFrame({
    'Reg.no':[201,402,237,204],
    'Department':['CSE','EEE','ECE','IT'],
    'Name':['Nethra','Priya','kavitha','Raj'],
    'Mark':[89,85,90,98]
})
print(structured_data)
```

	Reg.no	Department	Name	Mark
0	201	CSE	Nethra	89
1	402	EEE	Priya	85
2	237	ECE	kavitha	90
3	204	IT	Raj	98

```
] import pandas as pd
import matplotlib.pyplot as plt
data={"Name":["Nethra","Priya","Kavitha","Raj"],
      "Mark":[89,95,90,100]}
df=pd.DataFrame(data)
plt.bar(df["Name"],df["Mark"])
plt.xlabel("Name")
plt.ylabel("Mark")
plt.show()
```

